

BUIDL CTC 2026 FALL · DEFI

Kitty

Savings circles where every payment is proven, not promised.

Ethereum stablecoins · settled and credit-scored on Creditcoin · via the Attestcoin Protocol

PROBLEM

Hundreds of millions save in circles nobody can see.

- Chit funds, susu, tandas, chamas, stokvels: ten friends, one pot a month.
- They fail two ways: the treasurer runs, or a member stops paying and nobody outside the group ever knows.
- Ten years of perfect payments builds zero formal credit history.

TRUST TODAY

Who you have to trust

STEP	INFORMAL CIRCLE	“ROSCA ON-CHAIN” APPS	KITTY
Holds the money	Treasurer	One contract on one chain	Escrow vault on Ethereum
Knows who paid	Treasurer’s notebook	Same contract	Attestcoin-proven txs on Creditcoin
Decides the deadline	Treasurer	block.timestamp / admin	Attested source-chain block height
Picks the recipient	Treasurer / lottery	VRF oracle	Fixed order or by proven score
Credit history	None	None	Kitty Score, readable by any lender

SOLUTION

Money on Ethereum. Rules on Creditcoin. Proofs in between.

- Members pay into a minimal escrow vault on Sepolia.
- The attester network attests the block; one batch proof covers the whole round.
- KittyLedger verifies through precompile 0x0FD2, decodes and binds every payment, enforces deadlines from 0x0FD3.
- Payouts are proven back. “Paid” is never an operator’s word.
- Any member can prove a round from the browser: the Proof Builder serves CORS, the ledger doesn’t care who submits.

LIVE LOOP · PAY → ATTEST → PROVE → VERIFY → CLOSE → PAY OUT → PROVE BACK

The first circle, live: Delhi Chit Circle



CIRCLES

3

live · Creditcoin CC3 Testnet

PAYMENTS PROVEN

14

100% on time

BATCH PROOFS

4

14 tx through 0x0FD2

TUSD SETTLED

850

payouts proven back

ATTESTED SEPOLIA
BLOCK

11687780

latest · 0x0FD3

KITTYLEDGER

0xC2A1...F276

Creditcoin CC3 Testnet ·
Blockscout

● proven on time ● proven late ● paid · proof pending ● missed

— receives this round ✓ already received

3 members · 100 tUSD a round · rotation by Kitty Score · round 3 of 3

What the ledger actually checks

- Batch `verifyAndEmit`: up to 10 txs, one continuity proof, one call for payments pooled across every open circle, preflighted free with the view `verify`, 24% cheaper than singles on the live `0x0FD2`.
- `EvmV1Decoder`: receipt status 1, exactly one Contributed log, emitter = registered vault, tx.to = vault, tx.from = member, exact amount, current round.
- Query id = `keccak(chainKey || height || txIndex)`, same as `ASCBASE`; per-circle chain key validated with `get_chain_by_key`; vault allowlist keyed by chain.
- Deadlines from attested height: `is_height_attested(chainKey, deadline + 64)`; a miss records the attestation that proved it.
- The dashboard asks `0x0FD3` which attestation covers a payment: `find_lowest_attested_after`, `get_attestation_bounds`.
- Proven against the live precompile before deploying: a real Sepolia proof → `0x0FD2.verify = true`; tampered bytes and a wrong chain key `revert(pnpm verify:live)`.

src/asc/KittyLedger.sol · recordContributions

one precompile call

```
function recordContributions(
    uint64 chainKey,
    uint64[] calldata heights,
    bytes[] calldata encodedTxs,
    INativeQueryVerifier.MerkleProof[] calldata merkleProofs,
    INativeQueryVerifier.ContinuityProof calldata continuity
) external {
    bytes32[] memory queryIds = _prepareBatch(chainKey, heights, encodedTxs, merkleProofs);
    uint256 n = heights.length;

    bool ok = VERIFIER.verifyAndEmit(chainKey, heights, encodedTxs, merkleProofs, continuity);
    if (!ok) revert ProofRejected();

    uint64 lo = type(uint64).max;
    uint64 hi;
    for (uint256 i; i < n; ++i) {
        processedQueries[queryIds[i]] = true;
        _recordContribution(chainKey, queryIds[i], heights[i], encodedTxs[i]);
        if (heights[i] < lo) lo = heights[i];
        if (heights[i] > hi) hi = heights[i];
    }
    emit BatchVerified(chainKey, lo, hi, n);
}
```

Eight ways to cheat: six decoded rejections, two accepted by design

- Replay → QueryAlreadyProcessed
- Fake vault with a perfect event → WrongEmitter
- Same proof, chain key 3 → WrongChain
- Included but reverted tx → SourceTxFailed
- Paid after the deadline → accepted, flagged late, -20
- Steal the steward's key → NotOperator / OwnableUnauthorizedAccount / RoundStillOpenOnSource / VaultNotTrusted
- Fire the agent → accepted: the ledger checks the proof, not the caller
- Poison the reasoning → three invented figures stripped before display

Reviewed adversarially before submission: only owner-trusted vaults can feed the ledger, nobody can be penalised for a circle they never consented to, a 64-block grace window protects on-time payers from a hostile early close, and pots only go to members who paid this round.

An agent whose only power is proof

LAYER	HOLDS	WHAT IT DOES
1 · The ledger	final say	Verified proof, receipt status 1, trusted emitter, right sender and target, exact amount, current round, unseen query id. Any failure: nothing happens.
2 · Deterministic decisions	timing only	Prove now or wait for a fuller batch? Missing the grace window costs 120 points: urgency beats thrift. Decisions logged.
3 · Cited reasoning	none	Claude explains the log in plain language. Any sentence with an uncited or unverifiable figure is stripped before display. No API key: the deterministic sentence, identical behaviour.

THE STEWARD'S LAST THREE DECISIONS · RECORDED ON CC3 TESTNET

web/src/data/steward.sample.json · 15 entries

payout12 Sept, 07:26 UTC

paid circle 2 round 0: 250 tUSD to
0xB077B088E668386Bc57af87F175d250f459f0791

circleId=2round=0recipient=0xB077B088...0791

amount_tUSD=250

sepolia 0x19eb4a2c29...

confirm12 Sept, 07:34 UTC

payout for circle 2 round 0 proven back to Creditcoin

circleId=2round=0sourceHeight=11687449

creditcoin 0x0dac06f03c...sepolia 0x19eb4a2c29...

confirm12 Sept, 07:34 UTC

payout for circle 3 round 0 proven back to Creditcoin

circleId=3round=0sourceHeight=11687418

creditcoin 0x156b8935c7...sepolia 0x364b223a91...

KITTY SCORE → CREDIT

The score is used, not just displayed

- 500 base · +15 on time · −20 late · −120 missed · 300–850.
- KittyCreditLine on Creditcoin underwrites from the score alone: tier A borrows 100% of proven volume, B 50%, C 20%, D nothing.
- A soulbound Kitty Score badge renders live from the ledger, so a lender can read it on any explorer.
- Every input is a proven transaction or an attested deadline. This is the data Creditcoin was built to carry.

The first circle has already settled

- KittyLedger, KittyViewer, KittyUSD, KittyCreditLine and KittyBadge deployed to Creditcoin CC3 Testnet on 12 September 2026 (KittyLedger 0xC2A1...F276); its paired Sepolia vault 0xa27e...DA84 is trusted on the ledger; the first circle and all eight attack scenarios ran on the first ledger and vault.
- Delhi Chit Circle, three members, 100 tUSD a round, rotation by score: three Sepolia payments, two batch proofs verified by the live 0x0FD2, an early close, a 300 tUSD payout on Sepolia and its proof back to Creditcoin, all inside twenty minutes.
- Attack scenarios rerun against the real precompile: the forged chain key is rejected by 0x0FD2 itself. Every transaction is in docs/TESTNET_LOG.md; the hosted dashboard, steward log and attack lab read the same chain.

ACTION	CHAIN	TRANSACTION	GAS
createCircle · Delhi Chit Circle, 3 members, 100 tUSD, 200 blocks	Creditcoin	0x342f75d7...3cb89c	354,101
recordContributions round 0 · batch of 1 through 0x0FD2	Creditcoin	0xaf5a3477...3e3ec0	390,516
recordContributions round 0 · batch of 2 through 0x0FD2	Creditcoin	0x05ef192d...39c0ef	588,527
closeRound 0 · everyone paid, early close	Creditcoin	0xc9ca3583...4cd89c	341,740
payout round 0 · 300 tUSD to member 0	Sepolia	0xe8b660fa...626841	64,821
confirmPayout round 0 · proven back through 0x0FD2	Creditcoin	0x7bda53a5...3c0da9	372,484
deploy KittyLedger v2 · startHeight bounded by the attested frontier; a miss cites its attestation	Creditcoin	0x41753630...83ab68	4,889,709
deploy KittyLedger v3 · payments cannot predate the circle (the ledger read today)	Creditcoin	0x3053c63b...3c411c	4,901,416

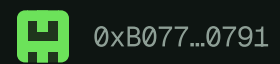
From circles to credit lines

- Score-gated circle sizes and seat bidding.
- Lender integrations on Creditcoin.
- Attestcoin writability for payouts once audited; Ethereum mainnet chainKey on CC3 mainnet.
- Pot cover through proven-event insurance.

TEAM

Prashant · solo builder

Foundry · TypeScript · React. Repo: <https://github.com/Prashant-thakur77/Kitty>. Contracts on Sepolia and Creditcoin CC3 Testnet; 162 Foundry tests · 29 agent tests · 8 attack scenarios end to end in CI; verified against the live Ox0FD2.



member 0 · first pot received

KittyLedger 0xC2A1...F276